(b) Explain with functional block diagram the excitation system of an alternator.

(c) Determine the K.E. stored by a 250 MVA 60 Hz two pole synchronous generator with an inertia constant $H$ of 5·4 MJ/MVA. Assume the machine is running steadily at synchronous speed with shaft input of 331100 hp. The electrical power developed suddenly changes from its normal value to a value of 200 MW. Determine the acceleration or deceleration of the rotor. If the acceleration computed for generator is constant for a period of 9 cycles. Determine the change in power angle in that period and the rpm at the end of 9 cycles.

**Unit-V**

5. (a) Define "Generation Shift" factors and "live outage distribution factors". What is the significance of these factors in contingency analysis?

(b) Write short note on power system state estimation.

(c) Define power system security. What are the operating states in a power system? Briefly discuss them in relation to security analysis.



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# 559112(24)

## M. E. / M. Tech. (First Semester) Examination, Nov.-Dec. 2015

**(Old Scheme)**

**(Electrical Engg. Branch)**

**(Specialization : Power Systems Engineering)**

## COMPUTER AIDED POWER SYSTEM ANALYSIS

***Time Allowed : Three hours***

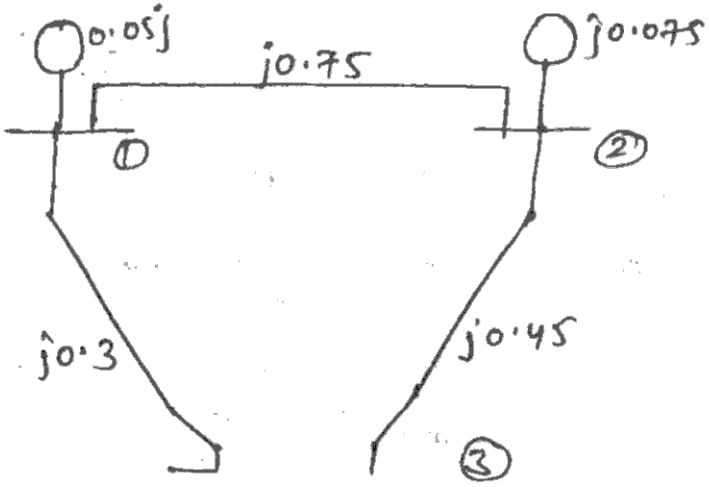
***Maximum Marks : 100***

***Minimum Pass Marks : 40***

*Note : Attempt any two parts from each unit. All questions carry equal marks.*

**Unit-I**

1. (a) Obtain the new bus impedance matrix for given figure.



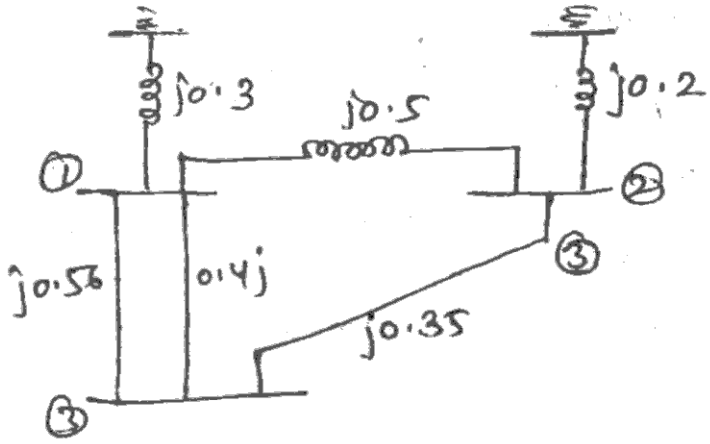

(b) The bus impedance matrix for the network shown in fig. is found to be :

$$Z_{bus} = \begin{bmatrix} j0\cdot183 & j0\cdot078 & j0\cdot141 \\ j0\cdot78 & j0\cdot148 & j0\cdot106 \\ j0\cdot141 & j0\cdot106 & j0\cdot267 \end{bmatrix}$$

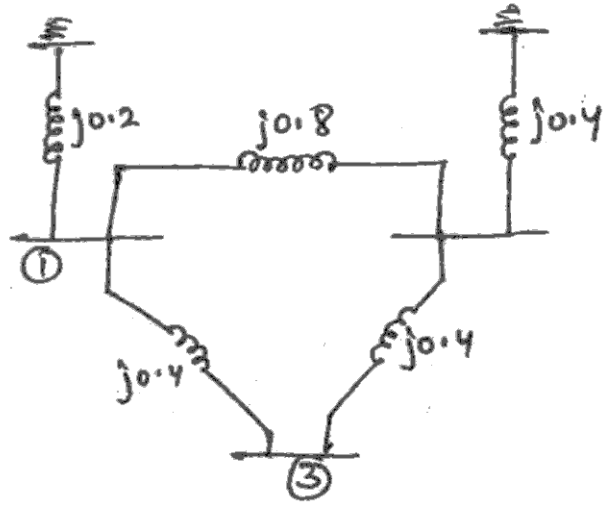
The line between bus 1 and 3 with impedance $Z_{13} = j0\cdot56$ is removed by the simultaneous opening of breakers at both ends of the line. Determine the new bus impedance matrix.








(c) Constuct the bus impedance matrix for the network shown in fig.



## Unit-II

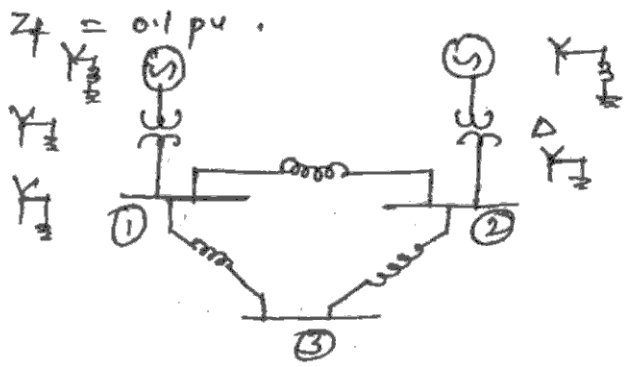
**2.** (a) The one-line diagram of a simple power system is shown in fig. The neutral of each generator is grounded through a current limiting reactor is 0·25/3 on a 100 MVA base. The system data expressed in per unit on a common 100 MVA base is tabulated below.

Determine the fault current for following fault :

(i) A balanced $3\phi$ fault at bus 3 through a fault impedance $Z_f = j0 \cdot 1$ pu.

(ii) A line to line fault at bus 3 through a fault impedance $Z_f = 0 \cdot 1$ pu.

(iii) A double line to ground fault occurs at bus 3 through a fault impedance $Z_f = 0 \cdot 1$ pu.








| Item | Base MVA | Voltage rating | $X_1$ | $X_2$ | $X_0$ |
|---|---|---|---|---|---|
| $G_1$ | 100 | 20 kV | 0·15 | 0·15 | 0·05 |
| $G_2$ | 100 | 20 kV | 0·15 | 0·15 | 0·05 |
| $T_1$ | 100 | 20/220 kV | 0·1 | 0·1 | 0·1 |
| $T_2$ | 100 | 20/220 kV | 0·1 | 0·1 | 0·1 |
| $L_{12}$ | 100 | 220 kV | 0·125 | 0·125 | 0·3 |
| $L_{13}$ | 100 | 220 kV | 0·15 | 0·15 | 0·35 |
| $L_{23}$ | 100 | 220 kV | 0·25 | 0·25 | 0·7125 |

(b) The per unit but impedance matrix for power system is given by :

$$Z_{bus} = j\begin{bmatrix} 0 \cdot 240 & 0 \cdot 140 & 0 \cdot 200 & 0 \cdot 200 \\ 0 \cdot 140 & 0 \cdot 2275 & 0 \cdot 175 & 0 \cdot 175 \\ 0 \cdot 200 & 0 \cdot 175 & 0 \cdot 310 & 0 \cdot 310 \\ 0 \cdot 200 & 0 \cdot 1750 & 0 \cdot 310 & 0 \cdot 500 \end{bmatrix}$$

(i) A bolted three-phase fault occurs at bus 4. Using the bus impedance matrix. Calculate the fault current bus voltage and line currents during fault.

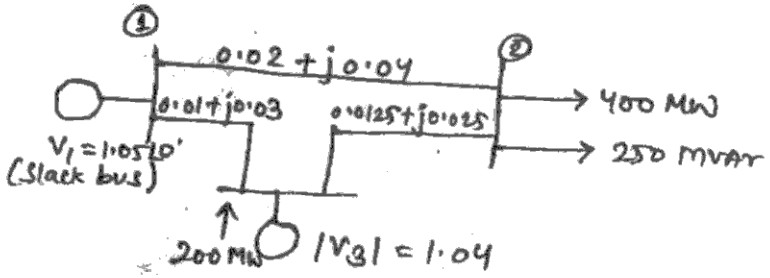
(ii) Repeat (i) for three phase fault at bus 2 with fault impedance of $Z_f = j0 \cdot 0225$.

(c) Develop sequence networks of a loaded synchronous generator from fundamentals and prove that

negative and zero sequence networks have no voltages.

## Unit-III

3. (a) Figure shows the one-line diagram of a simple three bus power system with generators at buses 1 and 3. The magnitude of voltages at bus 1 is adjusted to 1·05 pu. Voltage magnitude at but-3 is fixed at 1·04 pu with a real power generation of 200 MW. A load consisting of 400 MW and 250 MVAr is taken from bus-2. Line impedances are marked in per unit on base 100 mVA base and the line charging susceptances are neglected.



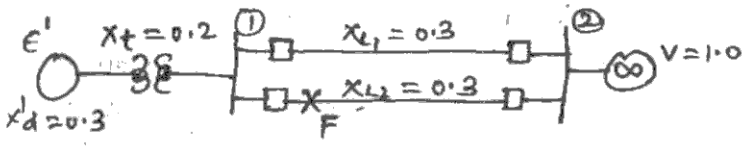

Obtained the power flow solution by the fast decoupled method for the system shown in figure.

(b) Derive the expression for Jacobian elements in Newton-Raphson method and state briefly how these can be simplified is Fast Developed technique.





(c) Develop a complete flow chart for the Gauss-Seidel load flow for a power system which has all the three types of buses.

## Unit-IV

4. (a) A 60 Hz synchronous generator having inertia constant $H = 5$ MJ/MVA and a direct axis transient reactance $X'd = 0 \cdot 3$ per unit is connected to an infinite bus through a purely reactive circuit as shown in fig. Reactances are marked on the diagram on a common system base. The generator is delivering real power unit $P_e = 0 \cdot 8$ per unit and $Q = 0 \cdot 074$ pu to the infinite bus at a voltage of $V = 1$ pu. A temporary three phase fault occurs at the sending end of the line at point $F$. When the fault is cleared, both lines are intact. Determine the critical clearing angle and the critical fault clearing time.